

Taunub: 2nd ATLAS circulation comments (EXOT-2025-06)

- CDS (v1.0): <https://cds.cern.ch/record/2954752>

[Information]

- Glance: https://atlas-glance.cern.ch/atlas/analysis/papers/details.php?ref_code=EXOT-2025-06
- INT note: <https://cds.cern.ch/record/2932059>

[editing]

- Overleaf: <https://www.overleaf.com/project/692ee49949631efd63ff5aea>
- git: <https://gitlab.cern.ch/atlas-physics-office/EXOT/ANA-EXOT-2025-06/ANA-EXOT-2025-06-PAPER>
- Paper draft: <https://cds.cern.ch/record/2952175>

[meeting]

- EB meeting: <https://indico.cern.ch/event/1653941/>
- PAM: TBA (<https://indico.cern.ch/category/2635/>)

[color]

Takuya

Jiaqi

Stergios

Junichi

Jan (comment-011)

Dear all,

thanks for the update. I have a few minor additional comments, but will then pass you on to Guillaume for the final checks (spokesperson deletage signoff).

Cheers, Jan

I. 163 is this "mu" reused later on? if not, we may also skip the definition to reduce confusion

>> Removed

I359 typo "injecte"

>> Fixed to injected

Figure 3 and similar captions: There s a statement missing about overflow contained in the right-most bin (I assume that s done). Also, we can skip "while the associated uncertainties are still shown correctly"

>> Added "The overflow entries are added to the rightmost bin." and removed "while..."

refs 16+17 need some {"u} umlauts

https://atlaspo.cern.ch/internal/reference_checker/ANA-EXOT-2025-06-PAPER/

>> Fixed

Caption of figures 5+: make sure to put the "(a)" or "(a,b)" etc ahead of the item being described, as for the prior figures. E.g. "at (a) $\beta_{L23} = 0.2$, (b) 1.0..."

>> Fixed

figure 10/ table 9: Sorry, I clearly had not yet given this proper attention. Can you expand both captions a bit to explain very clearly what is shown here? I now realise this is an alternative signal model that is NOT part of the paper. Say clearly what you mean with "this study". These figures will be standalone on our web page, so any relevant information needs to be put in here. Why would the visible cross section limit depend on the signal model? What you write "The visible cross section is defined as the cross section calculated with MadGraph, multiplied by the acceptance and selection." cannot be right: this is possibly the definition of the expected visible cross section, not the 95%CL limit on that quantity?

>> captions updated

https://gitlab.cern.ch/atlas-physics-office/EXOT/ANA-EXOT-2025-06/ANA-EXOT-2025-06-PAPER/-/blob/master/ANA-EXOT-2025-06-PAPER-auxmat.tex?ref_type=heads#L250

https://gitlab.cern.ch/atlas-physics-office/EXOT/ANA-EXOT-2025-06/ANA-EXOT-2025-06-PAPER/-/blob/master/ANA-EXOT-2025-06-PAPER-auxmat.tex?ref_type=heads#L260

Guillaume (comment-010) Replied

Dear Takuya, analysis team, editors

thanks for the answers. I don't have further comments from my side, but I noticed that in the pdf for the auxiliary figures 11 and 12 the caption have disappeared. I would also think that these figures should be labelled "ATLAS Simulation" instaed of "ATLAS"

cheers

Guillaume

Jan (comment-009) Replied

Dear authors,

Please focus on my first batch of comments to get the plots into their final shape as soon as possible for presentation at LHCP.

>> Done.

As next step, I now read the paper text and send comments - still on the on the draft dated 15th May 2026 10:44 v3.0 (taken from <https://cds.cern.ch/record/2952175>) as part of the PubCom final sign-off.

This PubCom reading is meant to bring a fresh look and assure uniformity across ATLAS papers.

When you produce an updated draft version, it would be helpful if you could provide a colorized difference between the current version and the new version. See information on how to do this at

<https://twiki.cern.ch/twiki/bin/view/AtlasProtected/PubComLaTeXDiff>.

You could upload this LaTeXdiff as an extra file in the CDS record of the working draft version.

>> Done

Please make sure you are using the latest version of the atlaslatex package. You can find the latest tarball at <https://twiki.cern.ch/twiki/bin/view/AtlasProtected/PubComLaTeX>.

>> Confirmed

You can also drop the cover pages.

>> Done

+++++

Congratulations on this nice analysis and the well-written paper!

L21 past tense showed a deviation - not true any more?

>> Now changed to “show”

L23 reference 12 is not very good - it's a web page. Cannot we have a publication here?

>> Updated it like this:

“Y. Amhis et al., Averages of b-hadron, c-hadron, and τ -lepton properties as of 2018, Eur. Phys. J. C 81 (2021) 226, the latest update is available at

<https://hflav-eos.web.cern.ch/hflav-eos/semi/ckm25/html/RDsDsstar/RDRDs.html>, arXiv: 1909.12524 [hep-ex]”

(some pheno papers refer to the latest HFLAV average like this)

L18-29 I don't have a complete suggestion, but I note these first 10 lines of the paper are very dense and technical. To me it would have made sense to start off why what are leptoquarks and why they may be interesting (refs 14-16 are definitely not the original idea of leptoquarks!), then bring the RD^* discrepancies. The technical discussion what model is target here (l26+) does not need to be here - your signal model section is very long, so no need to start that already here

>> Moved “This search focuses on the production of a vector leptoquark U_1 with quantum numbers $(\mathbf{3}, \mathbf{1}, \{2/3\})$, which correspond to the colour, weak, and hypercharge representations, respectively.” to Section 3.

We also added more references to original ideas of LQ, that are ...

<https://inspirehep.net/literature/230560>

<https://inspirehep.net/literature/235471>

<https://inspirehep.net/literature/92111>

<https://inspirehep.net/literature/89207>

L38 suggest to use the $\gtrsim$ symbol

>> Fixed

Figure 1: I'm not sure the terminology for b/s is clear - this is really a b-quark or a strange-quark? Also, the caption introduces a lot of new terms (soft jet, (non-)resonant signal) - this should not appear in the figure caption without being commented on in the text. Can the text refer to these two cases and introduce the terms, very briefly?

>> Updated the text and caption accordingly.

L57 is there a good reason to separate this CMS reference from the other references in the previous paragraph l30-41?

>> We added CMS taunu paper also to l.30, and kept l.57 as it is.

L85+1 just a curiosity: do you understand why the CKM matrix is applied here to the up-type quark? Usually we put these onto the down-type quark?

>> We followed the convention in <https://arxiv.org/pdf/2103.16558>

Added “The flavor basis of the quark fields is identified with the mass basis of the down-type quarks.”

L115 say missing transverse momentum (ETmiss) - not introduced yet?

>> We also use MET in l.171 before its definition in l.199. We added the definition in Section 1 i.e. “A search for leptiquarks is performed...missing transverse momentum, whose magnitude denoted by $\cancel{E}_T$ ”

L116-117 Depending on the analysis category defined later, events are required to satisfy either the E_T^{miss} trigger or single-lepton trigger. - I d drop that and explain later (this sentence doesn't give the reader much, it only confuses)

>> OK, this sentence was added to follow the comments during the ATLAS circulation, but happy to drop it

L120 probably and up to two partons (including b-quarks) ? Not very clear if exactly one b-quark required here

>> OK, updated to “up to two partons (including possible b-quarks)”

L123 Pythia8.308 (one dot less)

>> Fixed

L137 suggest to give the Sherpa version number used (or alternatively drop version number at other places, like l146)

>> Added

L149 EvtGen for all but Sherpa

>> Fixed

L152 actually the Pileup mode is different for rel 22 - see

https://gitlab.cern.ch/atlas-physics-office/atlaslatex/-/blob/master/template/MC23_snippets/Pileup.tex?ref_type=heads#L3 - feel free to shorten a little

>> Thank you. We misunderstood that mc20 uses the same pileup overlay method as mc16. We've updated the text as you suggest

L204 not so clear at this stage why you would want to have an electron or muon in your events. It would appear this discussion is maybe better for the next section?

>> OK. Removed

L216 you had already discussed the tight selection, can this be dropped/reduced?

>> OK. Removed

L219 are defined for lower LQ masses (1.5 TeV). - this sounds not quite right. Maybe you mean target signals with lower LQ masses (1.5 TeV). ?

>> OK. Updated

L222 why Due to the b-jet veto in this region, ? I d just drop this.

>> OK. Removed

Section 6 Event selection and 7 Background estimation - it actually appears that section 6 defines the SRs, while section 7 defines the CRs?

>> yes, but we think we can keep the current section titles

In any case, what s missing around l235 is a statement that you use the simulation to estimate backgrounds. You explain nicely how it s constrained in the CRs, but you fail to say that what is constrained is the MC sample normalisation

>> Added "The SM backgrounds are estimated using simulated samples."

L236-8 this is unclear - I would suggest to explain this properly - how can you get the fake tau rate from Z->ll studies? Almost the same text comes back at l302-306, but one still does not learn what was done really

>> Added more explanation in Section 8.

L247 improve the statistical precision. - how can removing events improve the statistical precision?

>> It is a left over from the version when we used SR1b only. Removed.

L251 drop while

>> OK

L252 it's unclear if the 90% purity is before or after relaxing these cuts or true for both. I would move the 90% purity to after discussing the full selection to avoid confusion

>> Moved it to the end of the paragraph

Does VRtop follow the relaxed kinematic requirements of CRtop or those of the SR?

>> Added "As with the CRTop regions, looser selection cuts for N_b , $p_t^{\tau_h}$, met , mt and $m_{b\tau_h}$ are required."

L274+ More citations would be nice here. I think the corrections applied are not interesting numbers to give, the uncertainty is more relevant.

>> We've added some additional references.

L283 what is pile-up subtraction?

>> We meant the jet-area correction. As a part of JES calibration, we just removed it

L295 maybe it would be good to add a few more recent ATLAS papers on this topic, e.g.

<https://atlas.web.cern.ch/Atlas/GROUPS/PHYSICS/PAPERS/PMGR-2021-01/>

<https://atlas.web.cern.ch/Atlas/GROUPS/PHYSICS/PAPERS/STDM-2018-43/>

(ref 78 is super old)

>> Thanks for suggestion. Added. We'll keep [78] given it mentions to W/Z+b simulation explicitly

L312 to be clear, these are Separate *background* normalisation factors?

>> Added

L314 is it a good idea to use lambda for normalisation factors, when one of the model parameters discussed in the introduction was using lambda (l38)?

>> The team and EB are more in favour of keeping lambda for that since it's one of the most common options (and other alternatives like κ and μ are also used for other things). So hopefully the subscripts/superscripts will distinguish them from the model parameter.

L321 for $\mu=1$ - this comes a bit late in the sentence? Suggest ... the expected contribution from destructive interference for $\mu=1$ amounts to

>> Fixed

L323 the total yield (BSM signal plus interference) - total yield should be SM+interference+BSM, not interference+BSM? Which statement are you trying to make?

>> Updated to “total BSM signal plus interference yield”

L324-327 suggestion (but please check): The analysis sensitivity is dominated by the SR1b regions. However, for larger values of μ a sizeable part of the signal populates regions without a reconstructed b-jet and the SR0b regions are sensitive to such U1 signals.

>> Looks good, adopted.

L327 make a paragraph break before For both μ , I d suggest to be clearer as For both the background-only and signal-plus-background fits

>> OK

L345+ not sure I fully understand: first you say with respect to the best-fit value of the signal strength μ . but then you say pseudo-experiments. With signal corresponding to $\mu = 1$ - can both be true at the same time?

>> Rephrased to “The impact of each systematic uncertainty on the analysis sensitivity is evaluated with respect to the best-fit value of the signal strength μ . This evaluation is performed using a dataset from pseudo-datasets constructed from the post-fit background expectation with an injected signal corresponding to $\mu=1$, and the impact is assessed on the fitted value of μ .”

L358 corresponding to parameter points for which $\mu = 1$. ? Not clear, can this just be dropped?

>> OK, removed

I wonder if one should say clearly that the analysis has no sensitivity to the B anomalies favoured region ?

>> We think this is something that readers can see for themselves. We prefer not to mention that explicitly.

I had already commented on the figures, nothing to add at this stage.

Conclusion section: this is a bit short overall, but ok for me if you want to keep it like it is.

>> We’ve decided to keep it as is, if there is no objection.

I notice you had some discussion in the intro (l36-41) that is not picked up again.

>> This discussion is added mostly in the intro to introduce the reader to the other related analyses that have been done. Then the complementarity with these searches is discussed again at the end of the Results section.

Table 9: the last lines is rally $m_{U1}=4$ TeV? Before the upper limit appeared to be 3 TeV.

>> Yes, it is. We didn't produce 3TeV signals for right-handed scenario...

References: https://atlaspo.cern.ch/internal/reference_checker/ANA-EXOT-2025-06-PAPER/

These look largely good. However the reference checker has identified a few cases where the journal capitalisation of the title differs from what you have, click on the Show no spaces no cases top left - can you check the title capitalisation vs. the journal-published version and correct where needed? The differences in journals and experiments/authors look overall incorrect, so leave as is.

E.g. refs 1, 2, 4 are affected

>> We tried to update them. Some items are marked by e.g. "Eur.Phys.J" v.s. "Eur. Phys. J" but can we keep them as they are?

Guillaume (comment-008) Replied

Dear editors, analysis team

thanks for this interesting analysis and paper.

To gain time, I am doing the deputy SP final sign off in parallel to Jan. I don't have further comments on the figures and you can find below few minor questions or suggestions.

They are based on draft version 3.0 dated 15 May 2026 10:44 from <https://cds.cern.ch/record/2952175>

cheers

Guillaume

- line 189 "The particle identification and isolation efficiencies in the simulation are corrected to match those observed in data." This seems to repeat what is said more in details lines 195-197. Is this line 189 needed ?

>> Indeed. We removed the line l.189.

- line 315: out of curiosity are the single top /ttbar ratio similar or different between CR and SR ? Did you try to check if changing the tW/ttbar interference would affect in a visible way the MC extrapolation from CR to SR ?

>> The ratio of single top/ttbar is basically the same between CR and SR within the MC stat uncertainty. (although the stats in SR1b-NonRes is very limited). We also checked the tW/ttbar interference with the alternative DS sample and the nominal DR sample. The difference is basically less than 10% and covers the difference between SR/CR.

- Fig 9 and 10 The bins with quoted acceptance*efficiency are a bit scattered especially at high mass with many empty bins. Just to be sure, you have only signal MC samples for the points with a reported value ?

>> Correct, all points where we have the signal points are shown in the plots.

- Fig 11 and 12 caption "toy" -> "pseudo-experiments"

>> Fixed

And the fact that pseudo-experiments are used for the main limits presented in the paper does not seem to be explicitly said. Is it worth to be added for instance in the paragraph starting line 356.

>> Added "where the distributions of the test statistic are evaluated using pseudo-experiments"

Jan (comment-007) Replied

Figure 2: There is space to make the legend text a bit later

The colour palette is very grey - is there still a chance to make it a bit lighter / less grey? I'm not insisting because it's quite late in the process. Maybe here just zooming a bit on the top panel (to have less Z+jets that has

It would also be nice to make the signal lines stand out a bit more, maybe they can be just a little brighter colour? (Unless a change in W+jets colour gets rid of this issue). Also, if you could choose a bit more different colours for the 3 TeV example signal and the prefit, that would be nice - or you use a different line style for example signals and prefit.

>> Colour of 1.5 TeV and 3 TeV signals has been changed to brighter red and magenta (colour code "2" and "6"). And line width is increased a bit. We use explicitly this palette as also discussed in the PAM since it's the ATLAS approved one for colour blindness, etc. Thus we feel that light pink is probably not optimal. About the zooming in we intentionally went so low to ensure that reader that all processes are present in all regions (so we tried to ensure that some Z+jets yields show up in the plot of the VRW-NR region). The rest of the line style choices look very bad with this linewidth, so they were disregarded.

Figure 3: I would drop the selection arrow that indicate actually no selection (2nd left, 3rd right)

>> Done. (We added the arrow for those plots by another comment's request before..Hope current decision would fix.)

Normally we put empty sub-captions below the figures (a), (b), and refer to these in the legend, see e.g. <https://arxiv.org/abs/2603.13113> Figure 3+ from page 20

>> Done

Overall it appears you went a bit too far by adding so many plots to the main body (we had zero for draft 1, now we have 16) - we're very long for a PLB and they will not like having so many plots. I would consider shifting a few less important plots to the auxiliary material, so they are not lost, but focus on key plots in the main body. E.g. CRTop-NonRes and VRTop-NonRes show almost no distribution (more than half of the plot is empty) and we have the integrated yields already on figure 2. Maybe you can bring this down to 4-6 plots and move the deselected plots to auxiliary material.

>> We moved CRTop and VRTop plots to aux materials and focused only on CRW and VRW in the main body.

The x-axes titles are rather complex - please try to increase their size as far as feasible so one can easily read these.

>> Done

For the y-axis we would here typically indicate the bin width *without changing anything on the plot itself*. Here it would be Events / 100 GeV for each plot

>> Done

The caption should also have a comment on the blue triangle (out of bounds points) and if overflow is included in the right-most bin (I guess it is)

>> Added "Bottom panels show the ratio of the observed data to the background prediction. Blue upward triangles indicate ratio values exceeding the displayed y-axis range, while the associated uncertainties are still shown correctly."

Figure 4: same comments as for figure 3. Clearly the Srs are a bit more interesting, but then having 1 variable per SR in the main body is maybe enough - I'd pick the one that has the best visual S-B discrimination and move the others to aux material. All distributions appear to be very flat - strange? Anyway, did you consider linear scale?

>> We moved MET distributions to aux materials and focus only on mass (Resonance) and mT (NonResonance) SRs. Log scale is preferred due to the low yields to visualise all or most of the background processes.

Figure 5: also here subcaptions, see figure 3.

>> Done

Move the ATLAS/lumi/couplings labels a bit down to avoid axes ticks overlaps

>> Done.

For the top figure I'd consider showing only half the numbers on the x-axis (the 1500 1550 1600 is very dense)

>> Done

Figure 6: also here subplots, see figure 3.

>> Done

Move the ATLAS/lumi/couplings labels a bit down to avoid axes ticks overlaps

>> Done. Also adjusted the legend.

Figure 11: Move the ATLAS/lumi/couplings labels a bit down to avoid axes ticks overlaps

>> Done.

Bertrand (comment-006)

Ref. [72] seems to be an internal note, could you please check if we are allowed to cite such material?

>> Indeed. We just removed the sentence related to the pileup reweighting uncertainty, since it is a minor source of the uncertainties.

Bertrand (comment-005)

Physics / clarity:

Title: Search for "a leptoquark" sounds to me as if there was one leptoquark to be found in the entire Run-2 dataset.

Would "Search for singly produced leptoquarks [...]" avoid that ambiguity?

>> We would also searched for t-channel exchange of a leptoquark. The title was modified based on the suggestion by Physics Coordinators at the first circulation. We would like to keep it as it is.

L262: "CRs using an electron or muon, by testing them in regions with a τ_{had} "

-> suggest "CRs using a τ_{had} and an electron or muon, by testing them in regions with a τ_{had} and no light lepton"

>> Adopted

L275: "The associated absolute uncertainty in the τ energy scale is smaller than 1.5%": Does the "absolute uncertainty" refer to a relative uncertainty on the "absolute" (MC-based) TES calibration, as opposed to a "relative" (data vs MC) in-situ TES calibration?

If it is really an absolute uncertainty, one would also need to know the "absolute" TES response values to infer a relative uncertainty (easier to interpret).

If that is a relative uncertainty, maybe drop "absolute" to avoid confusion.

>> The number is coming from this slides:

https://indico.cern.ch/event/1533397/contributions/6452412/attachments/3043941/5378050/25_04_02.pdf

Rephrased it to "The associated uncertainty in the absolute tau energy scale..."

Minor textual comments:

L97: τ -> τ -lepton

>> fixed

L114: trigger -> triggers

>> fixed

L140: NNLO was not defined, unlike LO and NLO

>> fixed

L168: suggest to drop "depending on the analysis region and purpose", to avoid repeating "analysis", and the purposes are clearly explained in the next sentence

>> adopted

L314-315: "the fit results are applied to SRs and VRs" -> suggest: "the fitted normalisation factors are applied in the SRs and VRs"

>> we also apply the nuisance parameters (not only normalisation factors). We'll keep the current description.

Figure 2 legend: for signal, you have e.g. "(1.5 TeV, ...)" while in Figure 4 it reads "(1.5 TeV)", suggest to drop the ", ..." from Figure 2

>> Updated the figure.

Figure 6 caption: at the 2σ level -> at the 1σ and 2σ levels (since you show both)

>> fixed

David (comment-004)

>> Thank you for the updates. We accept all of your proposals.

Robert (comment-003)

Thank you for this improved paper draft. All are comments were thoroughly addressed, so we have no further comments on behalf of the Slovak cluster (Bratislava and Kosice).

Bill Murray (Warwick) (comment-002)

Comments:

cross-section / cross section? You are not consistent.

> Changed all instances to cross section

Abstract:

This finishes " as a function of its mass at 95% confidence level."

The precise CL seems less interesting than the mass range

-> " for masses between 1500 GeV and 3000 GeV."

Same in the conclusions

> Added

l77 This is not a theory paper. How about we drop the kappas from this equation and just mention that 'terms introducing LQ couplings to the SM gauge-boson fields have been removed, i.e. this is a gauge model'?

>> We would like to keep it. It was added after long discussions and iterations between the analysis team, the EBs, and the ATLAS reviewers during the first circulation and at PAM. We do not think it should be overly simplified further just for the sake of simplicity.

The U_1 has variable indices, μ or $\mu\nu$..I am still struggling to work out why. The first term has both, but the next, which looks like a mass term, just has one. Maybe help the reader here?

> we changed it to:

"where $U_{1\mu}$ is the vector LQ field,

$U_{1\mu\nu} = D_\mu U_{1\nu} - D_\nu U_{1\mu}$ is its field-strength tensor, ..."

to make more clear

l319 The interference between the signal and the SM is considered by normalising the events by a factor of $\sqrt{\mu}$

This I find inelegant. How about

-> The contribution due to interference between the signal and the SM is scaled by a factor of $\sqrt{\mu}$

> Done

Jan (o.b.o. CERN EP reviewer) 

Dear all, the CERN EP reviewer has approved the document, but has left comments that should be considered and answered - I copy them below.
Thanks, Jan

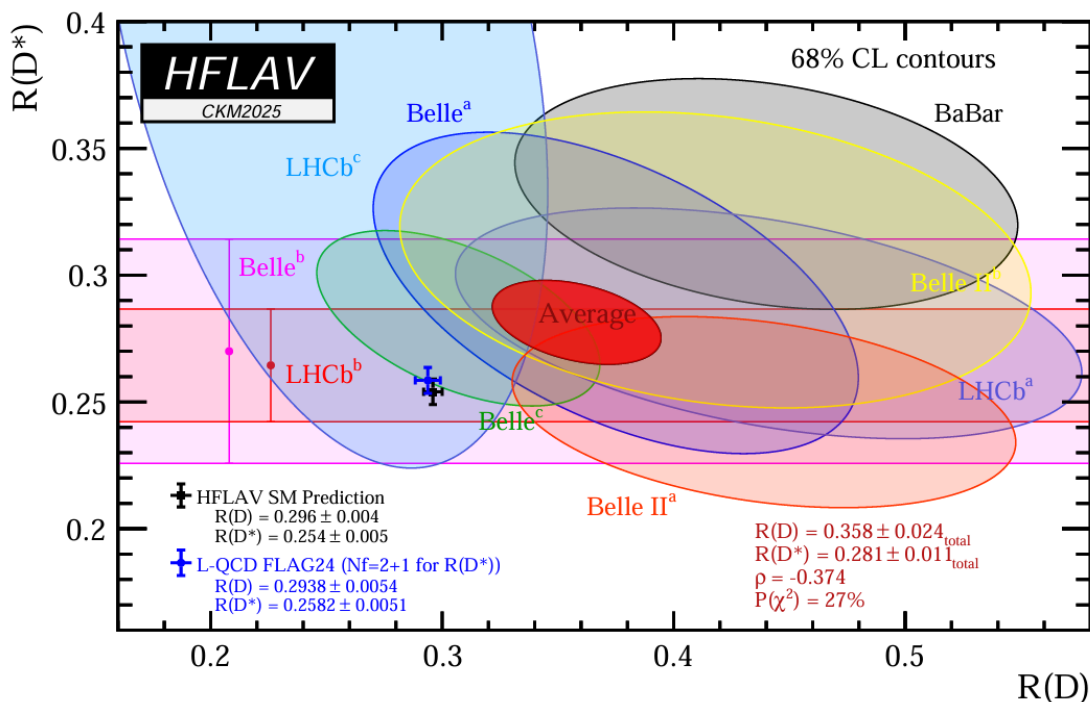
line 21: "results showed a deviation of about $3.5 - 3.8\sigma$ from the SM expectation on the ratio of the branching..."

I think this range makes some choices on the SM calculations available.
According to HFLAV, see for instance:

<https://hflav-eos.web.cern.ch/hflav-eos/semi/ckm25/html/RDsDsstar/RDRDs.html>

The range is something like from 1.4σ to 4.7σ . If you take the arithmetic average of the existing calculations correspond to about 3.8σ , but I'm not sure how to justify this approach.

> We've used $3.5 - 3.8\sigma$ since it corresponds to the deviations from the two SM calculations shown at the latest corresponding HFLAV figure (HFLAV SM pred & L-QCD FLAG24). We think that it's nicer to quote the average than giving the broad 1.4σ to 4.7σ range. So, we updated the text just to quote 3.8σ , and added "using HFLAV CKM2025 fit assumption" to make it clearer.



Taunub: 1st ATLAS circulation comments (EXOT-2025-06)

~~Christian on v1.2~~ answered  posted on CDS 

General:

You talk about the two benchmark scenarios for $\beta_{R/L} = 0, 1$ in sections 3 and 4, but then it's not mentioned again in the main body. The related studies then appear in the aux material. It's up to you if you prefer to keep the right-handed scenario in aux, but if you keep it, I would suggest mentioning it briefly in the discussion.

>> Added "In this analysis, the interpretation is performed in the pure left-handed scenario ($\beta_R=0$)." in section 3, and dropped right-handed signal description from section 4.

I would generally recommend referencing all material from the aux in the main body (also the tables, etc).

Abstract:

L7: I would remove the 'physics BSM targeting...' part to make the first sentence more punchy

>> Adopted

L10: you might want to add the Run 2 data taking period

>> Adopted

L15: Do you want to add a benchmark example of the parameter range you managed to exclude?

Body:

L77: I would define m_U and g_U again here to improve the reading flow (then also your reference to sec 3 in L40 fits again)

>> Added m_U here. We already have a definition of g_U in L.86

L84: I would clarify the benchmark scenario you are using more explicitly. If you keep both the left handed and the right handed, I would recommend that you mention both. If you decide to only show the left handed, say that you only probe $\beta_L=1$

>> See above

L85: you use R twice; am I understanding it correctly that its either $\beta_L=1$, $\beta_R=0$ OR $\beta_L=0$, $\beta_R=1$?

>> No, we have 2 scenarios with $\beta_L=1$ and $\beta_R=0$, and $\beta_L=\beta_R=1$.

Footnote 2: would it make sense to include this in the text?

L107-111: you might want to consider moving the trigger selection to the event preselection (section 5), but up to you.

>> No preference from our side, but we considered a comment from Warwick group.

L125: this is the last time you mention the right handed scenario -> consider removing if not discussed?

>> Dropped it

L153: what is the difference between τ_{had} and $\tau_{\text{had}}\text{s}$? It might be worth defining both?

>> Just a plural form. Rephrased to " τ_{had} candidates"

L159: 'background jets [59]' -> do you mean the mis-ID τ_{had} from q+g initiated jets?

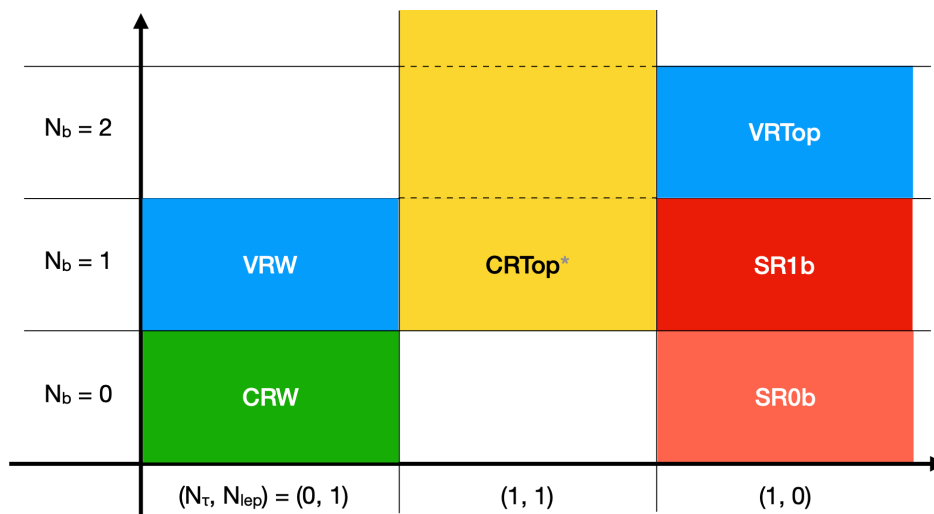
>> Rephrased to "hadron-jet background"

L204: it might be worth referring back to Fig1

>> Adopted

Sec7: I wonder if you have space for one of your VR/CR summary diagrams?

>> Like this? Added to the aux materials.



* For CRTop and VRTop, looser cuts for τ p_T , m_{Et} and m_T are applied for statistics

Fig2 legend and similar: $U_1(1.5)$ -> $U_1(1.5, \dots)$ to indicate that more parameters are varied as described in the legend? I assume adding all three would take too much space.

>> Done

Jan (PAM comments) - all answered

PAM meeting EXOT-2025-06 Tuesday March 3rd 2026, 11:00:

<https://indico.cern.ch/event/1657440/>

Search for leptoquarks in events with a hadronically-decaying lepton, b-jet, and missing transverse momentum using pp collisions at a centre-of-mass energy of 13 TeV with the ATLAS detector

GLANCE analysis link:

https://atlas-glance.cern.ch/atlas/analysis/papers/details.php?ref_code=EXOT-2025-06

Draft 1 link: <https://cds.cern.ch/record/2954752>

Please find below a short summary of the discussion of the PAM meeting.

The Editors and Editorial boards agreed to implement most of the CDS comments received by the Collaboration (6 people commenting)

Mostly items that required discussion and/or changes relative to the editors and editorial board proposal are mentioned below.

Changes to be implemented / checked agreed at the PAM are in bold.

Analysis presentation: Jiaqi Zang

Aurelio: p6 impact of this analysis, τ should not be a straight line

>> Plot not in the paper

Beta coupling assumptions are different from what is usually assumed in UV complete models (smaller β_{23}), are there other constraints on these?

Answer: no low energy constraint on β_{23} , already in paper

Aurelio: is really true, e.g. B_s mixing...

AI: authors to double-check if there are relevant references, see if coupling choices in MC section sufficiently motivated

>> Added the following sentence:

"Although $\beta_L \sim 0.1$ is often assumed in UV-complete models~\cite{Cornella:2021sby,Aebischer:2022oqe}, low-energy charged-current observables alone do not impose stringent constraints on this parameter.

Additional constraints arise from flavour-changing neutral current observables, in particular

$B_{s(d)} \rightarrow \bar{B}_{s(d)}$ mixing, as well as from LFU tests in τ decays.

However, these bounds are model dependent and may be affected by additional states in the UV theory~\cite{Aebischer:2022oqe}."

Jan: can we show at the end the complementarity better?

Answer: not easy as other analyses usually use low β_{23} , comments will have to remain qualitative

AI: make sure the results/conclusions are as good as possible

Aurelio: N-1 plots for main body or CR postfit

AI: we have 2-3 pages space left easily for PLB (15 pages is a good number, we re now at 12), so suggestion is to show around 4 plots in bkg determination to illustrate we control these background (no need to fill many pages, just to give the reader some useful information)

>> Added.

Aurelio: page 28 when discussing these plots, mention interplay with pair-production final states/analyses

>> Added a sentence related to the complementarity to the pair-production analyses.

Kerstin: p29 I recall these looked different before (res/nonres)

Answer: there was a change in signal samples, sensitivity initially hoped for didn't materialise

Kerstin: p20 excess coherent in e and μ is less likely to be statistical?

Answer: didn't find any detector related issues, pointed to various sliced regions

Jan: underestimate of $W+b$ cross section? Come back to this in EB talk

Collaboration comments: Flavia de Almeida Dias

S3-7 toys will be used, ranges a bit trimmed ok

Aurelio: can we put a b anomalies best fit line/area

Answer: ok, can be done

>> Added 1sigma/2sigma regions instead of the best fit line.

S8: ok

S9: changes ok, make sure you are consistent.

There was a suggestion to write Data/Pred. instead of Data/Bkg. - for team and EB to decide (I find other EXOT ATLAS papers with Data/Bkg.)

>> We would prefer to have it as Bkg. since it's less confusing and it's used like this in other ATLAS papers.

S10: add upper limit once as suggested, add 95% CL

>> Done.

Ticks around the frame (and move labels slightly inward)

>> Done.

S11 ok to leave unsmoothed (leave to decision by team/EB)

>> We prefer to keep it unsmoothed.

S12-15 very small effect, good to discuss as proposed, no need to go into too many details

S16-17 Jan: W+b underestimated?

Answer: probably an effect of tails and limited MC stat, 2sigma still ok

S18-20: Aurelio: this really looks like a SR; Jan: 20% improvement is significant

AI: it is decided to change this to a SR, this will need some rework of paper and changes/addition to the INT note; Exotics group and EB should sign off on the updated results. Drop b-jet from title. It will be good to motivate briefly why this region without b-jet requirement still has sensitivity to the targeted models

>> Done and added a sentence for the 0b SRs.

We can still continue with a 2nd circulation, stating the change in circulation message

>> Done

S21: can we double-check our cross section is right?

>> We checked the cross-section of btau signals generated with similar job options (to be confirmed by Tanaka-san?), which is consistent with the current ATLAS sample.

AI for exotics convenors that we are not missing something obvious in the cross section

>> Checked with CMS, but was hard to compare apples to apples.

S22: ok

S23: ok, suggest to state the typical range of signal widths in the simulation section and say that there is no narrow width approximation

>> Added the discussion about the width in Sec 3

S24: answer on CDS, no need to add plots or discuss (unless authors prefer)

S25: ok

S26: ok

S27: suggestion to cite ATLAS VH(bb)

<https://atlas.web.cern.ch/Atlas/GROUPS/PHYSICS/PAPERS/HIGG-2020-20/>

>> Added

Table 4 shows that the Sherpa sample (same as used by you) can easily be 30-50% off, so 30% is probably at the lower edge of a sensible systematics. (although there is no 1:1 translation of kinematics and your systematics is on the W+light/W+hf ratio).

>> It was shown that even pumping this number to 50% doesn't have any impact on the fit result/sensitivity so it was decided to stay with the current number.

S28: ok, no need to do excessive work

S29: authors will check again the highlighted A*e numbers

>> Done

S30 : ok, see if the general idea how SRs were chosen can be stated briefly in the paper, bkg stat uncertainty is still in good range of ~10%

>> Added a sentence for SR optimisation, changed 10% to 30% (more realistic number pre-fit)

S31 if sample exists and can be processed easily, authors can check the contribution

>> Studies were summarized in the Int note, the contamination was found to be small resulting to a few % change in the limits in mu (which translates to a smaller change in mass exclusion)

S32 ok (maybe citing the impact on the limit would have been a more obvious way to report this)

S33 extra tables and plots are good, as discussed already before. There is some space for the main body, so anything that can be seen as helping the reader significantly should be added. (but auxiliary material can also be used)

>> Moved some plots to the main body.

S34 just need to be clear what means NLO (nominal ttbar samples) and NNLO (needs ref)

>> Done.

The PAM is passed, congratulations . Please update to the latest atlaslatex package for the next draft.

We encourage the authors and EB to work together to address the CDS comments on Draft1 and the action items in a timely fashion. As there will be some changes in the analysis setup and results, a 2nd circulation is required. A short explanation of changes will need to be drafted and added to the circulation email if result changes are extensive, it will require a set of short slides.

Congratulations on this nice achievement!
Jan & Laura

Bertrand (Spokesperson deputy) - all answered

Physics / clarity:

Title and abstract: a centre-of-mass energy of $\rightarrow \sqrt{s} = \sqrt{s}$

>> Adopted

L33: "mostly model-independent" could be misleading, it could be interpreted as "model agnostic".

The searches are optimised for LQ signals, but the LQ model itself may have few free parameters.

Please rephrase.

>> Good point. Rephrased it to “In LQ pair production, the cross section depends only on the strong coupling constant at leading order, making the searches largely insensitive to the specific choice of the LQ coupling.”

L36: "high-mass": please define which mass

>> “...high-mass region of the final-state system”

Figure 1: Very minor question about the LQ propagator, it is shown as a dashed line (used for scalars?) while you study vector LQs so I was expecting a wiggly line?

I m not asking that you change the diagrams, it s just for my understanding.

>> Replaced it to the vector boson.

L105: Maybe remind the reader that you start at $M(U1)=1.5$ TeV because LQs are excluded up to $M(U1)=1.6$ TeV (mentioned in the introduction but it took me some time to make the connection)?

>> Adopted

L135-136: Since Athena Release 22, in the offline tau reconstruction, the classification of tracks as "tau tracks" is performed by a RNN track classifier that uses as input all the tracks associated with the tau (within $R=0.4$). Most often though, tau tracks are within $R=0.2$ of the tau (η, ϕ) axis (especially in your pt regime).

The geometric criterion R

>> OK... checked the PUB note again, and tried to improve the description.

“Candidate τ leptons are required to have $p_t > 20$ GeV and $|\eta| < 2.5$, excluding the crack region of the EM calorimeter $1.37 < |\eta| < 1.52$.

The vertex of a τ decay is determined from tracks within $\Delta R < 0.2$ of the axis of the seed jet.

Tracks associated with the τ candidate are then selected from all tracks associated with the seed jet based on geometric and impact-parameter criteria, and classified using an RNN-based track classifier~\cite{ATL-PHYS-PUB-2022-044}.

Candidates are required to have one or three tracks classified as τ tracks, with the sum of their charges equal to ± 1 .

Another RNN-based classifier is used to separate τ leptons from the background jets~\cite{ATL-PHYS-PUB-2022-044}, using tracks and calorimeter energy clusters associated with the seed jet, as well as several high-level discriminating variables.”

>> Also changed to τ_{had} and defined RNN

>> PUB note was checked again, and we tried to improve the description. We’ve also switched to the τ_{had} notation and defined RNN

L140 mentions that the loose RNN WP is used to count the number of taus, while L179 mentions that the tight RNN WP is used in the SR.

Is the loose WP only relevant in the context of tau veto, e.g. for the WCR and VRb definitions?

If so, this could be made a bit more explicit.

>> We use Loose WP to count the number of taus in the events. In the SR, we required $n_{\text{Tau}}=1$ and Tight WP. Added a sentence to clarify it.

Added “Different working points are used in this analysis depending on the analysis region and purpose. The “Loose” working point is used to count the number of τ leptons, and the “Tight” working point is employed for the τ lepton in the signal selection. “

L182-186: SR-Res targets $M(U1) \sim 1.5$ TeV. SR-NonRes targets the high m_T regime, but does that correspond to a particular range in $M(U1)$?

>> For non-resonance signals, signal strength scales to $\sim (g/M)^4$. We don't have any specific target mass range.

L194: Could you please explain why you use a single-lepton trigger instead of the MET trigger for the CRWs, where you require offline MET > 200 GeV?

In CRWs, are you still in the trigger turnon for MET ~ 200 GeV?

Or is it to be aligned with the CRTops where you cannot use the MET trigger due to the relaxed offline MET requirement?

>> Because it is a bit tricky in the muon channel, due to different mET definitions between online and offline. Yes, in CRW we apply the same mET cut as SR.

L198: It would be worth mentioning how the multijet background is estimated, and what uncertainty is assigned to it.

Is it negligible both in the SRs and in the CRs/VRs?

>> We confirmed it is negligibly small in all of SR/CR/VRs, after the Tight tau ID, high mET, and m_T requirements. In particular, the multijet background is estimated with recommended di-jets samples. And the source of multi-jets in SRs comes from a mis-measurement of high-energy hadronic jet, which leads to the fake MET. We therefore performed the cut on p_T and angular relation between leading jet and MET. This selection significantly drops the multi-jet and makes it negligible in our analysis, so we would prefer to not mention it explicitly.

L206 and L212: Regarding the choice of the leading jet as a proxy for the b-jet in the SR, is that for simplicity, or motivated by something else?

>> It's just for simplicity

L269: I assume this refers to the interference between signal and SM, not between s-channel and t-channel LQ exchange.

For the interference, is the event yield negative or positive, and how large is it compared to the "signal-only" yield?

Could that be mentioned in the text?

>> Indeed this refers to the interference between signal and SM. We found the negative interference effect is very small, after requiring a b-jet, because SM $W+b$ is suppressed by CKM matrix. Added the description in the introduction "An advantage of the $\tau\nu b$ channel is that the corresponding SM backgrounds are highly suppressed~\cite{Endo:2021lhi}. In the SM, the production of a $\tau\nu$ system with an associated b quark is CKM-suppressed, making the SM amplitude that interferes with new physics contribution highly reduced at the high-mass region."

L274: Would it be worth giving the fitted values and uncertainties for the background normalisation factors?

As the CRs are not 100% pure, we cannot directly read that from the ratio plot in Figure 2.

>> Added.

L275: From Figure 2 and Table 8, it seems that the excess is in VRb-NonRes, not VRb-Res.

>> Yes. Fixed.

Figure 2: Validation regions are called "VRb" in the text, but "VR1b" in the figure.

>> Changed to VRW (since the other one was promoted to a SR)

L300-301: Is there a particular reason to quote limits for $\beta_L^{23}=2.0$ and 0.6 , while these values are not the ones shown in Figure 3?

>> Limits are now quoted for the correct β_L^{23} values.

L308: The visible cross section was not defined.

>> Updated.

L312: According to section 3, signal samples are generated up to $M(U1)=3$ TeV, how do you extrapolate to 4 TeV?

The comment also applies to Table 9.

>> We have specific signal samples at $M(U1)=1.5, 2, 3, 4$ TeV and 5 TeV with $\{g_U=1.0$ and $\beta_L^{23}=0.2\}$ fixed for both $\beta_R^{33}=0/1$ scenarios. Already fixed the text.

Aux material:

Table 3: for the b-jet multiplicity, "(1)" should not be in bold

>> Fixed.

Table 5: $p_T(\tau)$ should not be in bold

>> Fixed.

Table 9: What SR-Res selection criteria are causing the drop in acc^*eff when $M(U1)$ increases?

>> The decrease in acc^*eff for the SR-Res categories when $M(U1)$ increases is mainly due to the change in the signal composition rather than a single selection threshold. As discussed in Section 3, the resonant contribution decreases with increasing $M(U1)$, while the non-resonant contribution becomes dominant. Since the resonant regions (basically SR1b-Res) are optimised for the low-mass resonant topology, a progressively smaller fraction of the total signal satisfies the SR-Res requirements at high $M(U1)$ and more events migrate to the SR-NonRes categories.

Interestingly, for SR-NonRes, acc^*eff seems almost unaffected by the LQ mass.

This behaviour is interesting and could be discussed in the main body of the paper.

>> I suppose it's simply because the resonance signals are "fading" as LQ mass increases. As our signal contains both resonance and non-resonance processes. And in the non-resonance SR, the signal kinematics distribution doesn't depend on leptoquark mass. Meanwhile, signal kinematics in resonance SR highly depend on leptoquark mass.

Minor, textual comments:

L28: consider moving the footnote to the main text

>> Done

L31: the LQ -> LQ

>> changed to "a" since it was reading better

L34: the single -> single

>> Fixed

L37: the LQ -> LQ

>> Fixed

L40: a similar -> similar

>> Fixed

L46: "An excess over the SM background has been searched for [...]" -> suggest: "A search for leptoquarks is performed [...]" or "This letter presents a search for leptoquarks [...]"

>> Adopted

L48: "the magnitude of which is denoted as ETmiss" could be omitted here, as it is also mentioned at L169

>> Adopted

L53: other channels -> in other channels

>> Dropped in new version

L84: Ref. -> Refs.

>> Fixed

L87: parameter -> parameters

>> Fixed

L110: the SM -> SM

>> Fixed

L127: single electron or single muon -> single-electron or single-muon (for consistency with L194)

>> Fixed

L131: is employed -> are employed

>> Kept "is" since it refers to data

L132: maybe not needed to introduce a "PV" acronym, it is only used twice

>> Acronym was dropped.

L142: "Loose" eVeto working point based on the RNN -> "Loose" RNN eVeto working point

>> Adopted

L150: the small -> small

>> Fixed

L179: "Tight" -> the "Tight"

>> Fixed

L186: tail regime -> regime

>> Fixed

L192: where the -> where

>> Fixed

L199: "for each SR using two control regions (CRWs), one for each SR": unclear whether this makes 2 or 4 CRs in total. Same phrasing used at L207.

>> It makes 2 CRs in total, we think that "one for each SR" clarifies it

L215: to prove the extrapolation "holds"?

>> Fixed

L220: reconstructed by -> computed with

>> Fixed

L230: comparison at -> comparison in

>> Fixed

L238: "corrected by ttbar samples" -> "corrected to match data efficiencies measured in ttbar events"

>> Fixed

L240: to take -> taking

>> Fixed

L243: "(PDFs)" has already been defined at L97

>> Fixed

L258: the SM -> SM

>> Fixed

L262: broken ref

>> Fixed

Figure 2 caption: batched -> hatched

>> Fixed

Figure 2 legend: Bkgd. -> Bkg

>> Fixed

L291 and L292: of the -> of

>> Fixed

L293: on the -> on

>> Fixed

L294: calculated by -> calculated with

>> Fixed

L299: the fixed value -> fixed values

>> Fixed

L303: to be "within"

>> Fixed

Tables 7 and 8: Expected background yields -> Observed yields and expected background yields

>> Fixed

Figure 5: the red arrow is not described in the caption

>> Fixed

Table 9: (TeV) and (fb) -> [TeV] and [fb], for consistency

>> Fixed

Table 9 caption: andSR -> and SR

>> Fixed

Slovak cluster - all answered

Major comments:

L122: On the previous line you mentioned that simulated events are weighted to reproduce the pile up distribution. However, on line 122 you mention that data are rescaled to achieve better agreement. Could you please clarify if you really also rescale data, and if so, what does it mean?

>> We follow the recommendation in

<https://twiki.cern.ch/twiki/bin/view/AtlasProtected/ExtendedPileupReweighting>

and the one described in this note (Section 2.2):

<https://cds.cern.ch/record/2014726/files/ATL-COM-SOFT-2015-119.pdf>

When we calculate the pileup weight, data μ value is first scaled by the DataScaleFactor, then MC distribution is weighted to that, so we can get better data/MC agreement on Nvtx distribution.

L231 - 233 you state the factors are consistent with 1 in all pT bins, do you see any shape effect for correction factors as a function of tau pT?

>> We refer to these slides:

https://indico.cern.ch/event/1406675/contributions/5916743/attachments/2839491/4962789/Slides_TauWG_2024_0417_ADM.pdf (1prong)

https://indico.cern.ch/event/1538698/contributions/6479548/attachments/3053342/5397780/Slides_TauWG_2025_0416.pdf (3prong)

L269: By the interference term of the signal sample, do you mean the interference between the diagrams shown in Fig.1(a) and Fig.1(b)? If so, it is not clear to us what is meant by normalising the events by a factor of $\sqrt{\mu}$, given that μ denotes the signal strength. Furthermore the ratio $\mu / \sqrt{\mu}$ depends on the MC x-section of the signal sample and would be different for different MC x-sections (even if the measured x-section was the same). Could you please explain why this behavior is desired?

>> No, it is the interference between signal and SM (off-diagonal term). It is described now. Anyway, we confirmed its impact is very small at the μ range we are interested in in this analysis.

L275: In the VRb-Res region, a deviation of about 2σ ... Is it really VRb-Res? From Fig.2 it looks like it is VRb-NonRes, isn't it?

>> It was a typo and fixed.

Minor comments:

L84-85: One may wonder why the couplings $\beta^{(31)}_L$ and $\beta^{(32)}_L$ are set to zero, i.e. why the leptoquark is assumed to couple exclusively to an antiquark and a lepton, and not to a quark and an antilepton? If this is indeed the intended assumption, a brief explanatory sentence would be desired.

>> Here we don't consider the 1st and 2nd gen leptons to focus on $b \rightarrow c \tau \nu$ transition motivated by R(D) anomaly. The quark and antilepton interactions are included in h.c.

L86: some explanation of Pati-Salam model would be useful.

>> We now removed these definitions.

L191-192: We suggest rephrasing the sentence Subdominant contribution as light-flavour jets. -> "A $t\bar{t}$ -bar event can mimic the signal if both the tau lepton and the bjet are misidentified as light-flavour jets. In contrast, the Wt process contributes only if the tau is misreconstructed as a light jet, while the b-jet is correctly identified."

>> We feel that the current description is adequate and we'd like to keep it as is.

L217-221: The definitions of the validation regions VRW and VRTop are well understood. However, the reason for introducing VRb, particularly its connection to b-jet physics, is not immediately apparent and would be useful to be clarified.

>> Added: "The \VRb regions are used to test the extrapolation of the background estimation from CRs with a b -jet veto to SRs requiring a b -jet. The \VRW and \VRTop regions probe the validity of the W +jets and $t\bar{t}$ background normalisations obtained in CRs using electron or muon, by testing them in regions with τ .".

L218: You use VRb here and in the appendix, while VR1b is used in Fig 2, please unify

>> Changed to VRW in the end (since the other VRW region was promoted to a SR).

L232: The sentence The correction is less than 1% with an absolute uncertainty of less than 1.5% is not entirely clear. Do you mean that the correction itself is smaller than 1% of the τ energy scale, and that its value is determined with an uncertainty of less than 1.5%?

>> Yes, for example 100GeV tau in the simulation is calibrated to 101 ± 1.5 GeV

https://indico.cern.ch/event/1533397/contributions/6452412/attachments/3043941/5378050/25_04_02.pdf

Rephrased a bit to "The correction applied to the simulated tau energy scale is typically below 1% of the reconstructed τ energy. The associated absolute uncertainty on the τ energy scale is smaller than 1.5%."

L262: Please fix the reference to TRExFitter

>> Fixed

L263-268: The background-only fit, performed in four single-bin control regions to constrain the normalisations of the W +jets and top-quark backgrounds, also includes additional background contributions (such as diboson and Z +jets processes) that are normalised using Monte Carlo predictions. Could you clarify whether the normalisations of these additional backgrounds are fixed to their MC expectations, or whether they are allowed to vary within their associated MC uncertainties? While this may be obvious to experienced TRExFitter users, it is not immediately clear to a general particle-physics audience.

>> They are allowed to vary within the uncertainties.

L.294-312: Parametric space of leptoquark processes is in general 4-dim. (M_{U1} , g_U , $\beta_L^{(23)}$, $\beta_R^{(33)}$) or at least 3-dim (M_{U1} , g_U , $\beta_L^{(23)}$) in pure left-handed model. At $\sqrt{s} = 13$ TeV we cannot expect M_{U1} to be higher than a few TeV, so your choices 1.5, 2, 2.5 and 3 are reasonable. However to expect that $\beta_L^{(23)}$ is higher than 1, means that leptoquark coupling to 2nd gen. anti-quarks is bigger than that to the 3rd generation. We would expect that $\beta_L^{(23)} < \beta_R^{(33)} = 1$. A brief clarification would be helpful.

>> We assume that $\beta_R^{(33)}$ in the comment refers to $\beta_R^{(33)}$.

In this work, the beta parameters are treated as effective flavour couplings without imposing a specific flavour symmetry or hierarchy. While certain UV-complete models may favour $\beta_L^{(23)} < 1$, we allow a broader parameter space to provide model-independent constraints, as discussed in the Introduction.

Table 5: τ $p_T > 200$ - it is written bold but it is the same condition as in the SR, i.e. it should not be in bold.

>> Fixed

Warwick - all answered

Physics, Signal Modelling

Appendix I of the INT note tries to reassure us that interference is basically negligible. But kinematic plots from https://indico.cern.ch/event/1353419/plots_largestat suggest the reverse - interference dominates over BSM. (They also look incorrect: LQ+interference is not equal to the sum of LQ and interference.) The paper has one sentence, saying it is accounted for, and another describing the approach in the fit, and that's it - no clue of the impact. Does this matter? We should tell the reader.

>> Actually, the interference effect is highly suppressed at high-mass by requiring a b-jet.

We added "An advantage of the $\tau\nu b$ channel is that the corresponding SM backgrounds are highly suppressed~\cite{Endo:2021lhi}. In the SM, the production of a $\tau\nu$ system with an associated b quark is CKM-suppressed, making the SM amplitude that interferes with new physics contribution reduced at the high-mass region. " to the introduction, and "The interference between the signal and the SM is considered by normalising the events by a factor of $\sqrt{\mu}$, but its impact to the signal yield is very small for the range of coupling values considered in this analysis. " in the Results.

The A^*E is a rapidly varying function of 3 parameters. We think we really should have a table of signal strengths (S+I) some S+I plots of the signal under a few different hypotheses, preferably in the paper but at least in the auxiliary. You do stress that a new region of space is being probed, but the reader does not know how that presents.

>> We have added some text to clarify further how these three parameters play out. There were a few plots of signal distributions for the different parameter points already shown in the PAM, but we think that the length of the letter doesn't permit to have additional tables in the main body, especially when the signal strength translates directly to cross-section limits (which are already shown in Figs 5 and 6).

Figure 7: There seem to be some complex patterns in this figure - or else stat/(sys?) errors are important. One caught our eyes: the signal at $m=2\text{TeV}$, $g_U=2$ and $\beta_{L^*23}=0.8$. The A^*E is lower than the points above and below it in β_{L^*23} by a noticeable amount. Are we confident we understand this? If it is just statistics, we should do some smoothing somewhere. If it's real, then do we know why?

>> Former numbers have missed the filter efficiency. The plots are updated and the difference is smaller. We also checked the kinematics of those points and found no specific reason for the feature. So we conclude that it's from the statistics.

Line 269: The interference term of the signal sample is considered by normalising the events by a factor of $\sqrt{\mu}$. We don't understand why this is valid.

>> This is the interference between signal and SM (off-diagonal term). If we assume the total amplitude is $A=A(\text{SM}) + \sqrt{\mu}A(\text{BSM})$, $|A|^2 = |A(\text{SM})|^2 + \mu|A(\text{BSM})|^2 + 2\sqrt{\mu}\text{Re}(A(\text{SM})A(\text{BSM}))$

According to previous ATLAS measurements [73], Ref 73 is about simulation setups, not ATLAS measurements, and we failed to find a recommendation for a 30% uncertainty in it. This is the second largest uncertainty, so perhaps more justification is needed?

>> We picked up 30% uncertainty from the PMG recommendation here:

https://twiki.cern.ch/twiki/bin/view/AtlasProtected/PmgWeakBosonProcesses#Normalisation_discrepancies_due
Indeed, the note describes the MC-to-MC comparisons (Section 5) and there are up-to 30% difference on W+heavy flavor fraction between different MC samples.

After the discussion at PAM, we updated the reference to H→bb measurement paper.

The width of these LQs is never mentioned. Does it matter?

>> The LQ width is properly calculated based on the LQ model we used in the MadGraph. The LQ width strongly depends on parameters, from about 10 GeV to several 100 GeV. Anyway, we take into account the LQ width in our MC samples.

The dependence of A^*E can be large. For example, at $M_U=1.5$ and $g_U=0.5$ it varies by a factor of more than 10 within the resonant mode. The validation plots we could find for this MC (<https://indico.cern.ch/event/1353419/>)

do not seem to explore the kinematic dependence on β^{23} , so we don't know how we can have confidence in this. In general, there could be islands in this plane, and it is unclear whether this is handled. [Ref Figure 7]

>> please find the plot in the slides at PAM

Analysis and Background Modelling

Fig 2: The VRW non-Res bin has a prediction of 38 data events (int note table 10.1) and a U1(3TeV) signal expectation of around 25 events. This seems like the best SR, but it is not used. Why??

>> After the discussion at PAM, we decided to change VRW to SR0b.

There is no mention of the pileup reweighting or scale factor corrections for the particle identification.

>> Added

Section 8: Line 264: template binned. Do you need to describe the B-only fit? The interesting fit for the measurement is the S+B. Can you present the fitted regions side by side? How many free-floating parameters? Are there 2 or 3 parameters for the background?

>> B-only and S+B fits show almost the same results (given the best fit $\mu \sim 0$). But we show both results, to discuss about the SM comparison first, and new physics interpretation afterwards.

Updated the fit model, and described the number of free floating parameters.

Line 193-194: The paper mentions that single-lepton triggers are used instead of the $E_{\text{t,miss}}$ to obtain some background samples. Is it OK to take data from a different trigger? Are there any issues with different kinematics between one trigger and the other, and do these triggers overlap?

>> We used mET trigger in SR and lepton triggers for CR/VR. Event orthogonality is ensured by the selection cuts. Using mEt trigger for events containing muons can cause the bias on mEt distribution since muon is not used for the mEt calculation at the trigger level. For this reason, we used the lepton trigger in CR/VR. We anyway require offline $mEt > 200\text{GeV}$ to take the similar topologies between SR and CR.

If we consider the trigger efficiency SF and its uncertainty correctly, there must be no problem to use the different triggers in SR and CR.

Line 275: In the VRb-Res region, a deviation of about 2sigma. Did you mean NonRes? post unblinded fit (fig 11.4 int) the error on the prediction in that bin shrinks and the ratio data/pred does not decrease. How significant is it then?

>> Yes. Typo fixed. About 2sigma is the post-fit number.

Section 4: You start by describing the trigger, what about CME, Luminosity and data periods, and pile up

>> Added a new section "Data and MC simulation"

Why do you rely on hybrid fit to estimate the systematic uncertainties? The covariance matrix from the final S+B fit will provide you with the full uncertainty list

>> We think estimating the impact of uncertainties on $\mu=1$ using Asimov dataset is a common strategy to evaluate the impact of systematic uncertainties on the signal strength. If we use the data directly, we would estimate the uncertainty on the best-fit $\mu \sim 0$, and this is not what we want to quote here.

Paper Structure

Our impression is that the theory is favoured over the actual analysis details. For example, the useful tables of definitions for control regions, number of expected events, etc., are relegated to auxiliary material, while there is 1 page discussion of the effective Lagrangian. I would have reversed that

>> Added more material in the main body related to the background modelling in CRs and VRs after discussion in the PAM

Lines 127-133 should be at the top of Section 3, MC simulation section and change the title of Section 3.

>> It is moved to the new section "Data and MC simulation"

Line 199: it is unclear if you have two normalisation parameters for W+jets or just two CRs. Would you add a summary table for the SR and CR and highlight the orthogonality?

>> We clarified 2 normalisation factors are used for W+jets and $t\bar{t}$ in Section 8.

"Separate normalisation factors are introduced for the resonance and non-resonance regions, resulting in four independent normalisation factors: two for W +jets (λ_{W}^{Res} and $\lambda_{W}^{\text{NonRes}}$) and two for the top-quark background (λ_{t}^{Res} and $\lambda_{t}^{\text{NonRes}}$). The top-quark normalisation factors are applied simultaneously to both the $t\bar{t}$ and single- t samples."

Rates: You provide eqn 2 explaining how the coupling depends on g and β for b decays, but don't do it for our search! Either equivalent formulae for the two signals (and interference) or a plot mapping it out to put with fig 7 (or both) would be most welcome

>> We have added some text (see above) about how our signal shapes vary according to the coupling parameters. This equation is basically used to calculate the regions favoured by the anomalies for each of the coupling parameter combination.

MC simulation section: Line 74-94 should go to the introduction, as it is a general.

>> we prefer to keep it as it is

Figures and Tables

Fig 3 Maybe Obs. -> "Obs. upper limit"? Also fig 4.

>> At PAM, we agreed to put "95% upper limit" outside the legend box, then Obs, Exp for each line.

Fig 4: We assume these curves are constructed using the efficiencies from fig 7. However, the jagged nature feels very non-physical. Can we not smooth them in some way?

>> At PAM, agreed to keep it as it is

Figure 6 shows the UL on the visible cross-section. If we only had one SR wouldn't this limit be independent of the mass? [Or rather extremely close to, as in the CRs the s/b is $O(1\%)$ but probably $M(U)$ dependent..] We are not sure what the purpose of this plot is. Table 9 seems more useful. Figure 6 (page 25): Why is the ratio of cross sections of the two β values significant?

>> If the Res/Non-Res ratio is not changed by adding the right-handed coupling, the visible cross-section upper limit must be constant, as you say. But as shown in Table 9, right-handed coupling can enhance the signals in NonRes region more (for SR-Res, on the other hand, $\sigma \cdot A^{\text{eff}}$ is not so changed by adding right-handed coupling). As a result, the limit with $|\beta_{R33}|=1$ is better than $|\beta_{R33}|=0$ at the highest mass region, where the sensitivity is dominated by SR-NonRes. That is what we wanted to discuss with this plot

Section 6: Can you present some kinematic plots to show the Data/MC agreement?

>> The kinematic plots were shown at the PAM and the N-1 plots were also added in the main body of the paper eventually.

Figure 5 (auxiliary material) (page 24): What is the significance of the red arrow?

>> Specified in the caption

Line 262: Broken reference; please also cite RooFit, Rootsta? Please add the TRexFier version.

Also, reference 47 looks like you are citing the ATLAS detector while you are citing the simulation infrastructure; It should be [47,48]

>> Fixed reference, too technical to add also the version, not done usually. Fix references as suggested

Table 7: What is the difference between < 0.01 and $'-'$?

>> $'-'$ for exactly 0. Harmonized to <0.01 .

Line by Line editorial comments

As you are going to submit your article to PLB, please use UK English: for example, program (6 times) programme.

>> Fixed

Use the present tense everywhere except the Data and MC section

>> Done (but if any other instance is found please let us know)

The title: "Search for leptoquarks in events with a hadronically-decaying tau lepton and b-jet, recoiling against missing transverse momentum using..."

>> We think that the title reads well already, and we would like to complicate it more

The title: Maybe you can add third generation

>> Same as above

Abstract: The first sentence in the abstract is 4 lines long and repeats the title! Maybe you can split it

Abstract: Does it really help to include so many numbers in the last couple of lines in the abstract? We got lost and would suggest dropping half the examples

>> Dropped them

Line 12: Do you need Two signal regions are considered , and maybe you can add the targeted mass range

>> Dropped them

Line 28: which is 3 and 1 in $(3, 1, \frac{2}{3})$ in bold? We assume it indicates they are vectors, but that is not explicitly defined

>> It refers to the different group representations and it's the notation used in the paper cited.

Line 30: Has the program been performed?! Or the search?

>> Changed to "developed"

Line 31: Reference ATLAS and CMS experiments

>> Done

Line 33: drop approximately

>> We would like to keep it since the previous limit was not exactly 1.6 TeV

Line 34: decaying to decaying into

>> Fixed

Line 35: assuming it was produced in association assuming associated production

>> Kept it as is since the meaning is the same

Line 38: Is $\sim$ favoured over $\gtrsim$?

>> We're not quite sure if they mean exactly the same thing, we would prefer it as is

Line 42: We would drop mentioned above and that the

Line 45: diagrams Feynman diagrams

>> Fixed

Line 46-48: Who performed the search? References?

>> Were given already in the previous paragraph

Line 65: toroids ranges between 2.0 and 6.0 T m across most of the detector : Is the T m a typo?

>> It's basically the product of T and m

Line 92: I would drop in general

>> general refers to the UV completion models, i.e. not "in general" as vague expression, decided to keep it

Line 94: Do you need thoroughly

>> We think yes, since it might be a very strong statement without it

Line 95-99: > 4 lines sentences!

>> Separated to 2 sentences

Line 151:

>> Comment probably for some other line / unspecified

Line 138: define RNN.

>> Defined

Line 145: define excluding crack region .

>> Defined

Line 158: Reference the overlap removal

>> No need for reference since it's described explicitly

Line 171-174: rewrite the sentence, as now sound

>> Kept as is

Line 190: I would replace is from .. by comes from W+jets and shorten the sentence.

>> Done

Line 77: its the field mass?

>> Yes, we think it's clear as is

Line 78: (Gell-Mann matrices) are the Gell-Mann matrices?

>> Yes

Line 111: boson bosons

>> It refers to the production, kept as is

Line 112: Please add the Sherpa generator

>> The versions are shown in the table below

Line 119: for Geant4, please use the atlas macro \GEANT

>> Done

Table 1: I would drop (full leptonic)

>> Done

Line 277: drop carefully, for example

>> Done

Line 321: "not covered by the previous tautau searches" -> "not covered by the previous tautau leptoquark searches"

>> Would like to keep it as it to be true for the general pp->tautau that just included LQ interpretations

Liverpool - all answered

Title/Abstract

L16 - $\beta^{\alpha\beta}$ should be defined (in some simple way) for the abstract

>> Dropped the description

Intro

L31 Assuming .. this sentence doesn't make sense

>> Changed "Assuming the LQ pair production" to "In LQ pair production".

[L31-32]: This depends if the t-channel diagram is included or not (qq LQLQ)

L31-32 means that for the LQ pair production, $g \rightarrow LQ LQ$ (see the 3rd term of the Lagrangian) is dominant. When we have two LQs in the t-channel diagrams, only one of LQs can be resonance (the other LQ is exchanged in t-channel). So such events are not selected with a LQ pair production search.

L46 You should make it clear at the start of this paragraph that you are talking about the present analysis. Maybe start with In the paper the results of a search ,,,

>> Merged this paragraph to the previous one.

L50 You need to say which channels it has the advantage over (suggest to add that in after "highly suppressed, compared to X"

>> It is now rephrased and focus on the discussion about the interference effect.

Signal Model and MC Simulation

L77 - M_{U1} has already been defined (albeit a forward reference to this sentence)

>> Removed it.

L86 - Pati-Salam model should have a reference

>> Removed these definitions for simplicity.

[L88] Might be worth mentioning that the choice of YM scenario affects only the production cross-section due to enhanced coupling with SM gauge bosons the kinematics of the LQ are identical.

We followed the previous LQ search papers' setting, that is, gluon $\rightarrow LQ LQ$ (gLQLQ coupling) is turned on. Indeed, there are some diagrams which have two LQs in the t-channel with the gLQLQ coupling. So, the

kinematics of tau etc can be changed when we turn off the gLQLQ ($\kappa=1$) in case of LQ mass $< \sim 3$ TeV. In the higher LQ mass, gLQLQ=ON does not affect our analysis.

L91 - It is not clear to me how this highlights the role of β_L^{23} in the $R(D^*)$ and I suggest this is spelt out more explicitly for the reader

>> Added "The term proportional to β_L^{23} is enhanced by the ratio V_{cs}/V_{cb} , implying that even moderate values of β_L^{23} can generate sizable contribution to $R(D)$ anomalies"

[L91] - Perhaps it should be clear that ν is the EW VEV, not some other VEV (i.e the breaking of the Pati-Salam scale or some other scale?)

>> OK

[L93] Maybe define what is the high β_L^{23} regime? I.e

>> Changed "regime" to "region"

L107 - The simulation of the interference term needs more details on the generator version, PS, PDF, tune, etc. Is it the same as the non-interference term? If so, this should be made clear.

>> Fixed

Tab 1 - NNLL and NNLO have not been defined

>> Added

Event Reconstruction and Preselection

[L135] using topological cell clusters [calibrated using a local hadronic calibration (LC)]. (add in the extra info)

>> Added

[L137] Candidate tau leptons if referring the τ_{had} candidates, but the phrasing makes it seem (to me) like leptonically decay tau candidates. Maybe just use the symbol (introduced after "Hadronically-decaying tau leptons (τ_{had})")?

>> Done

[L138] - Reference to 2022 Tau Reco paper, but the cuts defined in the text are using the 2019 Tau reco i.e tracks

>> Checked the PUB note again, and tried to improve the description.

[L153] perhaps better to say something like seeded from EMPflow cells or something similar to L135 for consistency?

>> We think it's ok to keep it as it is

Event Selections

L184 - Since the signal is only LO in QCD can we really trust it to model the distribution of additional jets and, in fact, also the $\min d\Phi(MET, Jet)$?

>> We limit the number of jets in our analysis.

L186 You don't need the word tail (implied by high m_T)

>> Done

Background Estimation

L195 You may want to say dominated by Z+jet events with Z->vv

>> We prefer to keep it as it is, since we also have Z->tautau component separately.

L196 - More details are needed for the fake background here for the reader to be confident this is correctly taken into account.

>> Since this is a minor source of background and it's taken directly from MC with a very conservative uncertainty, we feel that there are not more things to discuss here.

L196 Do you really need to say conservative?

>> We prefer to keep it

L199 - The CR (and VR) selections for both W and top should be summarised in a table, either together with the SRs in Tab 2 or separately since there are quite some changes that are difficult to follow and keep in mind in the text. Furthermore, couldn't the description of the electron and muon isolation here be included in the objects section? The extra selections here are essentially just implying "signal" electrons/muons

>> Due to the limit on number of pages for PLB, these tables were moved to the aux materials. It's a bit tricky to discuss the lepton selections earlier (e.g. in the object section) since there we define the looser criteria at preselection level (used for the region categorisation), which are tightened up later in the CRs. And since the SRs have a light lepton veto it's not very natural to discuss them earlier.

L205 This sentence seems to contradict the previous one. Maybe an and rather than full stop but it makes a long sentence

>> Swapped the sentences. Hm, let us keep the current sentences as they are, to avoid a long sentence

L207-L210 It is not clear now what selections are retained for these CRs. You should like you do for CRW state that all selections are the same and then list the differences (however as above, it would be easier for the reader if this was a table)

>> Added "All other selection criteria are kept the same as in the SRs."

L208 Maybe start a new paragraph for the CRTop control regions discussion

>> Adopted

L221 It would be worth explicitly stating that events with b-jets are vetoed in these VRs

>> We think it's enough with the descriptions in L218-219.

Systematic uncertainties

L224 - It would be good to quantify how much the analysis is dominated by the statistical uncertainties

>> The upper limit in mu deteriorates by 2% (1%) at 1.5TeV (2.0 TeV) for betaL23=0.2 and by 15% (9%) 1.5 TeV (3 TeV) for betaL23 = 2.2.

L243 - We should specify by what range the scale is varied. Same for top.

>> Done

L244 - On top of that -> Additionally

>> Fixed

[L259] - This is a little misleading I think. You state that the minor sources are "normalised" with a 30% uncertainty but in the results section the normalisation procedure is performed for the W+jets and top-quark backgrounds (understandably due to the definition of the CRs). I am assuming that the other backgrounds are allowed to float within their uncertainties (and are not specifically normalised). If that is the case, then I thought the standard approach was to apply the ATLAS measured cross-section uncertainty as the "theory" uncertainty? (and not an arbitrary 30%?) with the standard reco/detector uncertainties applied?

>> Already discussed at PAM. We prefer to keep the current 30% uncertainty since even the uncertainty of 50% won't have any impact on the fit results.

Results

L262 - Broken reference

>> Fixed

L265 - The fact that it is a single-bin cut-and-count fit should be mentioned earlier on in the introduction to frame the paper e.g. so the reader doesn't expect to see distributions of the final discriminant in the background section.

>> Added that

L269 - How do you deal with the fact that the interference contribution may be negative?

>> Negative interference effects are taken into account but signal strength (fit/model ratio) is limited to be positive. Anyway we found the interference effect is quite marginal and do not change the limits/results.

L275 - You talk here about the 2 sigma deviation in the VRb-Res but, from Fig 2, isn't it in fact the VR1b-NonRes? Also, do I see correctly here that the post-fit results actually describe this region less well than the pre-fit prediction. Was this investigated?

>> Yes, fixed it to VRb-NonRes.

We checked it carefully, and concluded it is a statistical fluctuation. Details can be found in Appendix G of the INT note. The excess is localized in a single mET bin. If we exclude that bin, it is relaxed to ~1sigma level. These events are not localized in a specific (eta, phi) region. No difference in e and mu channels. We also checked the region with looser mEt and mT cuts and observed good agreement.

L302 - You discuss the shaded area here but don't make the obvious comment that you exclude all the shaded regions at least in the parameter space presented in the figures? Unless this is not the case and instead the plots should be interpreted as moving "towards" sensitivity to the grey areas. Considering the very first lines of the introduction implies that the B anomalies were a motivation for this paper you should make the statement that the models presented do not offer an explanation for the anomalies at least in a certain region of phase space?

>> The curves shown correspond to 95% CL upper limits, such that the excluded region lies above the limit curve. In the parameter space displayed, the grey shaded region lies below the observed limits and is therefore not excluded. But this doesn't mean that the B-anomalies are not a motivation for the search since some of the couplings used are the ones preferred by the anomalies. It's just that in the end the analysis was not sensitive enough to exclude the regions. Added some additional motivation in the intro to clarify.

Figure 3 You should explain why you show results only up to 3 TeV even though that is where you obtain the best limit. Perhaps it's simply due to you not having generated samples but that should be said here.

>> Added: "The lower $\sqrt{s}$ boundary is chosen to cover the region around the current exclusion limits from LQ pair-production searches. At higher masses ($\sqrt{s} > 3\text{ TeV}$), the signal is dominated by the non-resonant contribution, and both the cross section and kinematic distributions depend on the coupling-to-mass ratio rather than the explicit mass value. Therefore, 3.0 TeV is taken as the upper boundary of the scan."

Conclusions

You need to repeat what this analysis says about the B anomalies. If you are not confident enough to mention it you should de-emphasize it from the Introduction, where it reads as the main motivation of the paper

>> It's already evident from the plots that the analysis was not sensitive enough to exclude the region favoured by the B-anomalies so no additional statement can be made apart from that. Of course we can write this explicitly in the conclusion, but it might seem weird.

Lailin (PubComm) - all answered

L16-18: Is it possible to briefly elaborate the physics meaning of the beta parameters, otherwise it's a bit hard to appreciate for the general readers?

>> Dropped the description

L27: I think a brief explanation of the features of leptoquark would be useful as for the introduction. Here you assume everyone knows what leptoquark means.

>> Adopted

L45: probably you can mention again the RD anomaly requirements in L29 to highlight the motivation more explicitly?

>> Adopted

L83: is this a repetition of L81?

>> Indeed. Removed the first part

L86: probably the Pati-Salam model is not clear to everyone?

>> Actually, we don't use the word "Pati-Salam model" afterwards, so simply we removed these definitions.

L95-99: the sentence is a bit too long without any pause?

>> Split to 2 sentences

L99-100: I assume with some MadGraph syntax one can separate out the s-channel contribution, or based on the truth τ invariant mass distribution one can roughly estimate the contribution of the resonant and non-resonant part?

>> If we check the truth-origin of tau lepton, we can separate s- and t-channels explicitly. Now we rephrased it to "All relevant production diagrams are considered in the ME calculation, including both s -channel (resonant) and t -channel (non-resonant) topologies." as suggested by Aurelio

L105: It seems to me the model is a bit complex with a few parameters. I think it would help a lot if you can either show some representative signal kinematic distributions with some typical sets of parameters (can be even in the Aux for example), or explain briefly how these parameters would affect the signal modeling. At least I find it hard to imagine the impact of the beta parameters. And what's the intrinsic width of the LQ?

>> To address this point we've added a more concise discussion (in Sec. 3) of how these parameters affect the signal. Basically, $m(U1)$ controls the relative importance of the resonant and non-resonant production, g_U sets the overall signal normalisation and β_L^{23} modifies the flavour structure and the fraction of signal events containing a final-state b -quark. We also describe now the corresponding intrinsic width behaviour and provide a representative example (in which the width is $\sim 7\%$). Also since the N-1 plots are part of the main body now they also contain representative kinematic distributions for the signal for different masses and couplings.

L107-109: How the combination is done exactly? The Interference term and the signal should have different scaling effects on the coupling parameters?

>> It is scaled by $\sqrt{\mu}$ and added to the BSM-only sample, as described in Section 8. We found the effect of the interference term is quite marginal, at the coupling scale we are looking at. Discussion about the interference effect is added in the introduction.

L171: electron or muon ?

>> It's here and indeed it's not quite clear:

"Events must contain at least one hadronically-decaying τ lepton, electron or muon, and at least one jet."

I would suggest either removing it or change to "zero or one electron or muon candidate".

L178: which exact triggers used? A bit contradictory to L127? OK, later I realize that lepton triggers are used for CRs. It might be good to make it clearer along L127?

>> OK, adopted

Sec 5: The selections seem to be quite tight. Can you comment on the typical signal acceptance times efficiency? It would be useful to add such information in the text.

>> The selection of the regions is motivated by a sensitivity-based optimisation (it's a high- p_T search after all). The acc x eff numbers are given already in the Aux material, and probably commenting on them in the main body wouldn't add much to the picture.

L252: Why take the difference between NNLO, which is assumed to be more accurate, and NLO as an uncertainty?

>> The $t\bar{t}$ sample is produced in NLO but then its cross section is normalised to the NNLO prediction. The uncertainty is just to cover for the shape difference between NLO and NNLO. Clarified it further in the text.

L262: What s the observable used in the fit? From Fig 2 it seems only the event yields are used for all SRs and CRs. Please make it more explicit in the text. Have you tried using shapes or kinematic distributions in the SRs for the fit, which I assume should give better sensitivity than the simple counting?

>> Added "The fit is performed using the observed event yields in each region, i.e. a single-bin likelihood is constructed for all CRs and SRs." The low statistics in the SRs doesn't help in fitting shapes there.

L267: What s the relative contribution of the resonant (non-resonant) signals in the two SRs?

>> It depends on the mass point. As illustrated in Fig.2, at 1.5TeV we see more events in SR-Res, while at 3TeV more signals in SR-NonRes.

Fig 2: It looks you have more signal yields in VRW regions than in the SRs? It would be useful to comment on this in the text.

>> At PAM, we agreed to change VRW as SR

L269: What s the typical size of the interference compared to the signal yield?

>> It is very small. Almost no impact on the final results.

L284: Asimov is a bit of jargon?

>> Changed to pseudo-experiments (as done in other papers)

Fig 3: I m curious about the relative contribution of the Res and NonRes regions to the sensitivity. Do you happen to have individual expected limits that can be added to the limit plots?

>> We have Res-only and NonRes-only limits in Fig.8 (aux material)

General: After the event selection description in Sec 5 or in Sec 6, I d expect to see some data/MC kinematic distributions to have a concrete feeling how the SRs/CRs look like. While the only data/MC plot is in Fig 2 and only one bin for each region. It would be great to move Fig 5 from Aux, or something like this, to the main body.

>> N-1 plots moved to the main body.

General: It seems to be that the current analysis is not optimal, at least for the following reasons: 1) you have extremely low signal acceptance times efficiency, from Tab 9 of Aux; 2) you have very low event yields in the SRs and lower signal yields compared to some other regions; 3) only simple counting with event yields used in the fit. Of course I might be wrong, but it would be helpful if some statements can be added to the text where appropriate to justify the choices explicitly.

>> at PAM, agreed it's ok if we briefly describe how the SR is optimized and MC stat uncertainty on bkg is 10%-sh

References:

Please fix issues identified by the reference checker (it may give some false positives):

https://atlaspo.cern.ch/internal/reference_checker/ANA-EXOT-2025-06-PAPER/

>> Done

Aurelio (PC) - all answered

General comments:

I understand this search is cut&count, but it d be nice to at least include some signal vs bkg kinematic distributions at the preselection level that help motivate the SR cuts.

>> The post-fit N-1 plots in SRs are found in Aux materials. We avoided to put them in the main body due to the limited number of Figures for PLB.

The SR cuts are very tight, leaving very limited MC statistics (as can be seen from Fig. 5 in the aux material). How were they optimized? Was there a requirement on a maximum MC statistical error? And was the optimization done on a background sample different than the one used to perform the statistical analysis to avoid a bias (analogous to the overtraining in MVA based analyses)?

>> In the SR optimization, we scanned the thresholds of kinematic variables with the guard requiring the number of total bkg yield to be >2.0. It was a bit arbitrary, but we found that the requirement can avoid the impact from the negative event weights for W+jets MC. Eventually we found the MC stat uncertainty ~10% on the SR bkg yield.

Please double-check what requirements should be marked as bold in Tables 3-6. E.g., in Table 5 I think the $p_{T\tau} > 200$ should NOT be in bold.

>> We checked tables and fixed it.

Did you check the signal contamination from $\tau\tau(+b)$ events with $\beta_{33L}=1$? They can also give a $\tau+b+ET_{miss}$ signature e.g. from a tau that is not identified or decaying leptonically and the lepton is missed.

>> The $b\tau\tau$ signals has been processed by ATLAS official AOD samples. It turns out that the $\tau\tau+b$ events could enter our SR1b to some extend. But the final impact on the $\beta_{23/gU}$ limit is sizable (~5%). So we would not consider it at this stage.

Detailed comments physics/clarity:

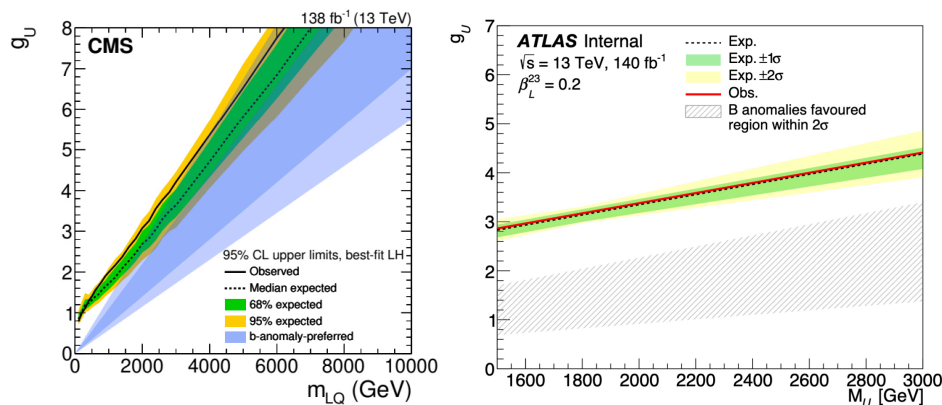
L51: are you trying to argue that the $\tau\nu b$ final state has more suppressed backgrounds than $\tau\tau b$? I wouldn't think that's the case. In the case of $\tau\nu b$, the W coupling to the b is CKM suppressed, but W+b-jets is easily produced through $g \rightarrow b\bar{b}$ splitting. I think a potential advantage is that the (negative) interference with the SM background must be smaller compared to the $\tau\tau(+b)$ channel? Was this checked? If so, I think that's something worthwhile mentioning.

>> True. What we wanted to emphasize is that $g \rightarrow Wb$ is highly suppressed compared with $q \rightarrow Wq$, which is not the case for Z+j, but indeed W+(g **$\rightarrow$** bb) was not considered in that

discussion (though we restrict the number of b-jets==1). Anyway, as you say, the point here is the interference effect is much smaller by requiring b-jet for W+jets. Yes, we confirmed the interference at high-mT region can be negligibly small after the b-jet requirement. We updated the text.

L53: The last sentence seems to be a bit disconnected from the previous discussion. You want to say that, in contrast with this analysis, the CMS reinterpretation didn't make a b-jet requirement, which significantly improves the S/B. BTW, how does the sensitivity of this search compare to the CMS reinterpretation?

>> This is a bit tricky to compare the results with CMS reinterpretation since we don't know their exact value of $\beta_{L^{23}}$. If we suppose they use the best-fit value (~ 0.2), and compare the observed limit with our results of $\beta_{L^{23}}=0.2$. Unfortunately, our limit is a bit worse than CMS. Maybe we can do the rough check with CMS event selection and see how the significance would change.



L114: What single top backgrounds are considered?

>> All s-, t- and Wt channels are considered. The Wt channel is the main contribution to our analysis.

L117: This is hard to imagine. How $WW \rightarrow \tau\nu$, $WZ \rightarrow \tau\nu b\bar{c}$, $ZZ \rightarrow \tau\tau b\bar{c}$ would not be more relevant than the fully-leptonic modes you are listing?

>> We require $m_T > 200$ GeV. Semi-leptonic diboson events do not lead very high m_{Et} . At the first round of the analysis, we also produced semi-leptonic diboson samples and found they are negligible.

L247: The 30% a-priori seems quite optimistic, given the extreme kinematic cuts that are being applied, compared to any ATLAS measurement. Also, where in the cited PUB can this 30% be found?

>> We followed the PMG recommendation (10-30%) and took the most conservative option, 30%. It says Section 5 of this note is the reference of this recommendation.

https://twiki.cern.ch/twiki/bin/view/AtlasProtected/PmgWeakBosonProcesses#Normalisation_discrepancies_due

L252: This is unclear. I assume you are referring to the application or not of the NNLO reweighting to $t\bar{t}b\bar{b}$? What nominal prediction is being used, w/ or w/o the NNLO reweighting? Please clarify.

>> Nominal prediction is not using the NNLO reweighting.

L257: What was the outcome of that check? How well did the yields agree? If $\sim 100\%$, that's a fair rather than conservative uncertainty.

>> As described in Appendix D of INT note, we observed 5-60% mismodeling of Z+fake tau MC. We compared various uncertainties on this (e.g. 50%), but the results are not changed at all since Z+jets contribution to our SR is very small. So we took the conservative option, 100%.

L259: What makes you believe these are conservative uncertainties given the disagreement seen in VR1b-NonRes? You are probing a quite extreme kinematic regime and it's not obvious these small backgrounds are predicted within 30%.

>> Those backgrounds are selected with real tau. So we think 30% is large enough to cover the difference between prediction and data.

L268: Make explain what background normalizations are considered, to what processes they are applied (e.g., is the Top normalization applied to $t\bar{t}b$ +single top?), and what are the measured values of these normalisations.

>> Updated the description and added the numbers.

Fig. 2: VRW-NonRes has a large signal contamination. This is expected since most of the NonRes signal doesn't have a b-jet within acceptance. Did you consider using this VR as a SR. What would be the impact on the sensitivity?

>> Agreed at PAM to change VRW as SR.

On the other hand, VR1b-NonRes shows a significant non-closure. How do you explain the reasonable agreement in VR1b-Res and the disagreement in VR1b-NonRes?

Did you check the different W+jets fractions (W +light, $W^{+>=1c}$ and $W^{+>=1b}$) in the different CR/VR/SRs? Did you relax cuts on the VR1b-NonRes to try to identify what kinematic distribution is responsible for the disagreement? Why shouldn't this non-closure be propagated as an uncertainty to the corresponding SR?

>> We checked it carefully, and concluded it is a statistical fluctuation. Details can be found in Appendix G of the INT note. The excess is localized in a single mET bin. If we exclude that bin, it is relaxed to $\sim 1\sigma$ level. These events are not localized in a specific (η , ϕ) region. No difference in e and mu channels. We also checked the region with looser mET and mT cuts and observed good agreement.

L297: Is this the difference in observed limits? 10-20% is not that small, in which direction it goes? Are the asymptotic limits more conservative or aggressive?

>> updated all results with Toy now

L305: Can we report the actual value of β_{23L} assumed by the tautau analysis? I couldn't find in the paper what value of β_{23L} was used to derive limits on gU vs mU1 (assuming $\beta_{23L}=1$). Are you sure it's not $\beta_{23L}=0$?

>> Tautau analysis used the signal samples produced with $\beta_{23L}=0$. If we understand it correctly, they assumed their signal strength does not so depend on β_{23L} , and compared the results with the B-anomaly favored region for $0.06 < \beta_{23L} < 0.16$. That's why it is tricky for us to make a fair comparison ...

L307: The complementarity is not in probing lower values of GU, but probing higher values of β_{23L} ?

>> tautau search has lower sensitivity on β_{23L} , while taunu has larger sensitivity on that. On the other hand, both final states are sensitive to gU. That's why we emphasize the higher value of β_{23L} .

L308: You didn't define visible cross-section.

>> Updated.

Textual comments:

It's odd not to see used the τ_{had} symbol throughout the paper. I think this should be corrected, since tau leptons can also decay leptonically.

>> Fixed

Title: a leptokuark ; a b -jet ; at $\sqrt{s}=13$ TeV (can make it a bit shorter)

>> Adopted.

L15: for the benchmark U1

>> Fixed.

L31: should also cite searches for vector LQ pair production in $b\tau/\tau\nu$ final states

>> Added.

L38: we use m rather than M to denote mass (apply throughout)

>> Done

L43: mention the assumption is that the coupling is suppressed because of the cross-generational coupling, involving a 2nd generation quark and 3rd generation lepton. But for an explanation of the $R(D^{(*)})$ anomaly this coupling must be non-zero, and thus it's well motivated to consider it.

>> Added a sentence there to motivate it even more

L44: for consistency with L42 you should call it $\tau\nu_{\mu}(+b)$

>> Fixed.

L45: illustrative Feynman diagrams for the signals targeted by this analysis

>> Fixed.

Fig; 1, caption: Illustrative Feynman diagrams ; exchange of a leptoquark ; with a soft

>> Fixed.

L47: a jet containing a b -hadron (b -jet)

>> Fixed.

L48: no need to define $E_{T\text{miss}}$ here, done later

>> Removed.

L73: Avoid excessive capitalization in section headings: e.g. Signal Model and Monte Carlo Simulation -> Signal model and Monte Carlo simulation (apply elsewhere)

>> Fixed.

L84: I understand you are assuming $\beta_{13L}=0$? If so, set it accordingly

>> We checked it again and found that it is set to $0.2\beta_{23L}$. However, we confirmed its contribution to $\tau\nu_{\mu}$ signals is quite marginal.

L87: parameters

>> Fixed.

L92: UV-complete models

>> Fixed.

L94: not explored thoroughly

>> Fixed.

L95-99: too long sentence, please split in two

>> Fixed.

L99-101: this sentence reads a bit odd just say that all relevant production diagrams are considered in the ME calculation.

>> Adopted.

L102-104: this is out of place, since you are discussing the generator level. I would suggest to move it when you introduce GEANT. You can then explain that fastsim is used for signal, but it should be clear that it is only used for the calorimeter simulation. The rest is simulated with GEANT.

>> Adopted.

L119: \textsc{Geant4}

>> Fixed.

Table 1:

Remove (full leptonic) . That information is already given in the text

>> Removed.

L127-133: cite DQ paper and PV reconstruction. Here you can find the recommended references:

<https://twiki.cern.ch/twiki/bin/viewauth/AtlasProtected/PubComUsefulBibTex>

>> Added.

L134: should introduce tau_had

>> Done.

L142: this is jargon, please rephrase as electrons misidentified as tau-lepton candidates

>> Rephrased

L154: I believe it is more important to say for what pT/eta cuts we apply JVT cuts than the actual working point used.

>> Done.

L165: remove using standard ATLAS-Run 2 calibrations

>> Done.

L174: end the paragraph indicating the above requirements constitute the event preselection .

>> Added “The requirements above define the event preselection, which serves as the starting point for the subsequent event categorisation and statistical analysis.”

L192: not all taus and b-jets are reconstructed as light jets, so this should be rephrased to avoid confusion

>> Updated slightly:

“Subdominant contributions arise from top-quark decays producing τ leptons (τ or $\bar{\tau}$), b or $\bar{b}$ quarks and W bosons ($W \rightarrow \tau \nu$ or $W \rightarrow b \bar{b}$), in cases where the τ leptons and b quarks are **misreconstructed** as light-flavour jets.”

L218: In Fig. 2 you VRb is called VR1b. Since it is dominated by W+jets, why not calling it VRWb?

>> Changed it to VRW (since the other one was promoted to a SR)

L239: cite also light jet calibration

>> Added.